

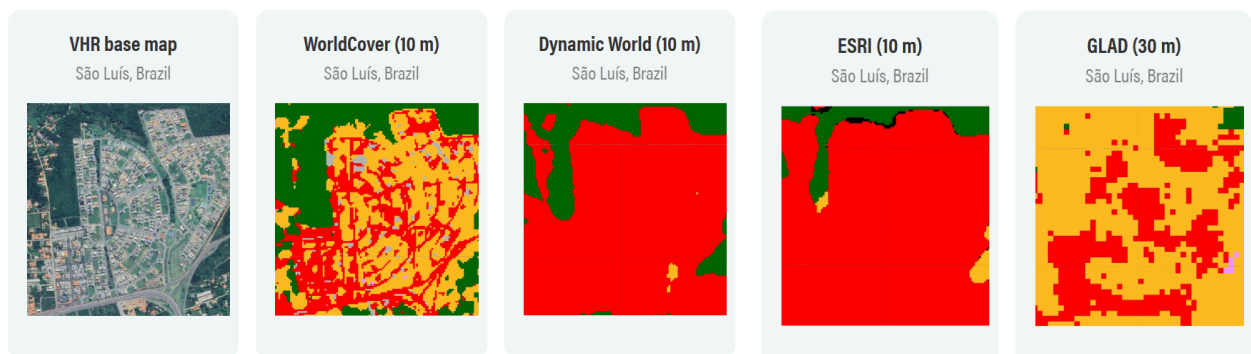
Addressing The Heterogeneity Problem with Automated Land Cover Classification

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Land cover is one of the most used tools in geospatial analysis. It has applications in urban planning, agriculture, risk analysis, climate change, disaster relief, and many other domains. Land cover assigns each pixel of a raster (a georeferenced grid of equally sized cells) to a class, which ranges from building, vegetation, road, bareland, or any of the other hundreds of classifications and subclassifications.

The most accurate way to map Land Cover is to hand-label each pixel, but this method is extremely time-intensive; a 1024 by 1024 pixel tile can take upwards of 2.5 hours to manually label, and at higher spatial resolutions, the labor requirements do not scale to the large swaths of land typically used in land cover analyses. Many deep learning models have been developed to classify land cover at a global scale using satellite imagery, significantly reducing the labor required to produce accurate and timely land cover maps.

However, a common issue these models share is accurately mapping heterogeneous urban areas. Detailed shapes and entire land cover classes are easily lost, especially at lower resolutions (Land & Carbon Lab). This is where high-resolution mapping comes in. By providing more detailed examples to a model, mapping densely packed regions can be more precise, as the model can learn the spatial context of the land and its surroundings more effectively.



Approach

Dataset: Open Earth Map

The dataset used for this model comes from a geospatial machine-learning organization called Open Earth Map (Xia et al., 2023). The dataset was compiled from a variety of high-resolution (.25m- .5m) remote sensing images of geographically diverse cities. A key limitation of this dataset, and land-cover classification more broadly, is the restricted number of semantic classes used to represent complex scenes. With only nine labels, certain objects become ambiguous. An example of this can be found in the test_models notebook (Mahe_15), where Unet misclassifies a boat as a building.

After storing the dataset, the first data preprocessing step was to create a class that could dynamically load in file paths for a raw image and its corresponding land cover label mask. The dataset was divided into training, testing, and validation sets. However, many of the raw images were missing from the dataset because they originated from a different high-resolution city dataset, known as the xBD dataset (Gupta et al., 2019), which is used for analyzing cities affected by natural disasters. I decided to forego incorporating these images because, during the training of my baseline model, the loss metrics I used stabilized in the training dataset, and the validation loss converged in the later epochs, indicating that the model had sufficient data and adding more would not significantly improve performance. After creating file path strings for the remaining image-label pairs, I set up a TensorFlow dataset generator function that loaded the raw TIF files as 512x512-pixel tensors, resized with Lanczos interpolation to preserve image clarity with minimal artifacts. Most images were either 1024x1024 pixels or 1000x1000 pixels, but there were outliers, such as the Paris images, which were 650x650 pixels (see test_models.ipynb for examples). The TIFs were loaded with RGB bands for the raw images, and a 9-class integer band for the labels. These pairs were loaded as tuples, allowing them to be passed into the models simultaneously.

Methodology: Unet and DeepLabV3+

The first model I trained was a classic U-Net architecture. The model consists of three main components: an encoder that extracts features from the input images, a bottleneck that captures high-level representations, and a decoder that generates the predicted label masks. The encoder is composed of four convolutional blocks, each containing a pair of convolution layers followed by batch normalization and ReLU activations, and ending with a max-pooling operation for downsampling. After the bottleneck block, the decoder upsamples the feature maps using transpose convolutions, mirroring the encoder structure. Skip connections concatenate feature maps from the encoder blocks to the corresponding decoder layers, preserving spatial information and enabling more accurate boundary reconstruction.

The U-Net model provided a strong baseline; however, object boundaries remained blurry and pixel-level misclassification was common, motivating the use of a model better suited to capturing broader spatial context. DeepLabV3+ is a state-of-the-art semantic segmentation architecture that combines multi-scale context modeling with precise boundary recovery. Like U-Net, it follows an encoder-decoder design, but enhances the encoder with atrous (dilated) convolutions to expand the receptive field without reducing spatial resolution.

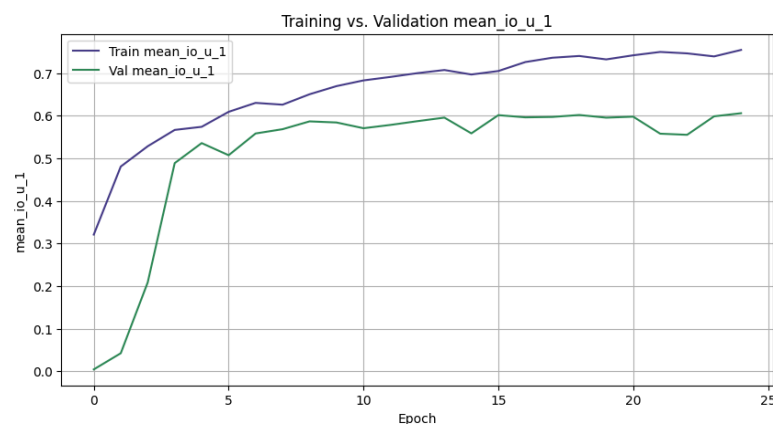
For this model, a pretrained ResNet50 backbone was used as the encoder. Trained on ImageNet, the backbone provides robust low-level feature extraction and boundary awareness, while excluding the classification head allows the network to be fine-tuned for satellite imagery. On top of the encoder, Atrous Spatial Pyramid Pooling (ASPP) captures contextual information at multiple scales through parallel dilated convolutions with varying dilation rates and global image-level pooling. The decoder then upsamples the high-level features and fuses them with low-level encoder features to recover fine spatial detail and sharpen class boundaries. This architecture effectively balances global semantic understanding with accurate localization, making it well-suited for heterogeneous land-cover classification.

Analysis and Results

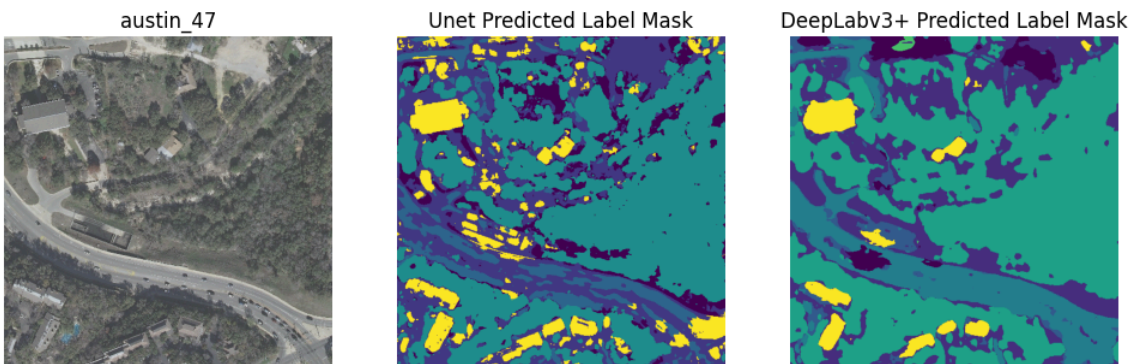
Quantitative analysis: Key metrics

Both models were primarily evaluated using the Mean Intersection over Union (mIoU) metric, as this metric assigns equal weight to each class and directly reflects how well predicted regions align with ground-truth labels. The U-Net model provided a strong baseline; however, both the Mean Intersection over Union metric and visual inspection revealed clear limitations. While training, the validation and training loss curves converged and plateaued early, indicating limited generalization. Prediction masks showed blurred boundaries and frequent pixel-level noise, especially in heterogeneous areas.

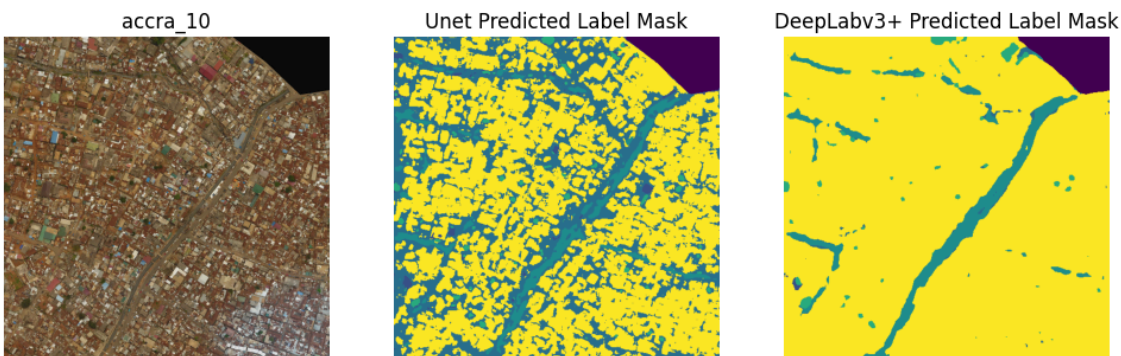
DeepLabV3+ exhibited rapid early gains in Mean IoU, outperforming the Unet in 4 epochs. The strong early performance reflects the benefits of the pre-trained ResNet50 encoder, rather than training weights from scratch. It's worth noting that I let the DeepLabV3+ train for ten more epochs than the Unet; this provided a marginal benefit in training performance, but plateaued in validation performance, indicating that the model had reached its peak performance with the available data, and likely overfitted to the training data because of the gap between the two curves. The metrics Dice Coefficient and Sparse Categorical Accuracy were also used in modeling, confirming the same conclusions. Plots of these metrics are present in their respective model notebooks. From left to right, the plots below represent Unet Mean IOU, then DeepLabV3+ Mean IOU.



Qualitative Analysis: Visual Inspection



Out of the two models, DeepLabV3+ created cleaner, more consistent shapes, with objects like large buildings and roads. As shown in the Austin example, Unet was able to get the broad classifications right, but when at closer levels, it often mixed classes into the bigger shapes. This is where DeepLabV3+'s spatial context comes into play; the larger kernel sizes enable a better representation of each object's surroundings and its edges.



This doesn't come with drawbacks, though, at very granular levels, DeepLabV3+ seems to overgeneralize its label masks as depicted in the Accra example below, where Unet was able to pick up on the developed space between the small houses, while DeepLabV3+ classified most of the land as a building.

¹ These visualizations lack a ground truth label mask because the Open Earth dataset did not have labels for its test set, more comparisons of the models' predictions can be found in the test_models notebook

Conclusion

Discussion & Interpretation

Increasing spatial resolution significantly improved land cover classification accuracy in heterogeneous urban landscapes, particularly when utilizing the DeepLabV3+ architecture to incorporate more spatial context. This allows for better analysis in Urban contexts, especially where hand-labeling costs valuable time. There are drawbacks to this approach; each tile was roughly 26.2 to 52.4 square kilometers of space. Mapping larger areas with this tiled approach could lead to a patchy map, as spatial information is not shared between tiles. Acquiring .25m to .5m resolution imagery does not scale to large areas either. These models were also trained specifically on heterogeneous urban landscapes, and would likely tank in performance for more homogenous images, especially with the Unet's tendency to speckle in random classifications.

Next Steps

Both models struggled to generate refined boundaries on objects, which, for most use cases, isn't a major bottleneck as long as general accuracy is high, but the next step to improve accuracy would be to learn finer edges. PointRend uses a rendering-based approach to upsample, creating cleaner edges (Kirillov et al., 2020).

While having the three color bands is useful for identifying differences in land cover classes, different reflectances could greatly improve the distinction between classes. For example, a Near-Infrared (NIR) band can greatly help in distinguishing between 'vegetation' classes, such as bareland and forest.

For implementation on actual mapping software, a sliding window approach could be used to mitigate the patchiness of the classifications. This method allows the edge pixels to be classified multiple times from different spatial contexts, reducing edge artifacts and creating smooth transitions between tiles.

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